Ethon – The Pointer Particle

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written in the formal posture of a doctoral thesis
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Abstract

This paper specifies the ethon as a real pointer particle of the theory. At the first layer of interpretation, the ethon is a single-particle formulation of interactions among the six quark flavors: the six flavors are represented as six directional arguments inside one real particle, and their readable interaction is represented as three-axis fluctuation. Each direction also carries a ternary interaction sign in $\{-1, 0, 1\}$, giving 3^6 external flavor-interaction signatures before rotation, polarity, and bonding data are read. It is a general meson implementation obtained by assigning boundary-free operational parameters to Yukawa's meson idea: not a replacement for the pion, but a purer meson whose role is to point from an unread relation to a stable readable entity before any particular physical boundary, mass, charge state, decay channel, or detector realization is imposed. The ethon has three transition axes: compression–exposition, upward–downward localization, and stabilization–turnover. Its fluctuation is simply rotation in this three-axis frame; no ten-dimensional or superstring-like ambient space is required by the core implementation. As a free functor it remains inert and has no up–down polarity. When an entity is placed in the T-direction, hidden polarity is born: the S-face turns toward the T-direction and the N-pole points toward S. This motion is not directly observable. When an entity is also placed in the S-direction, the ethon enters constant-speed self-rotation. The governing law is the Ihara equation, written around the oriented $T \to S$ premise–stabilization arrow. It gives the general meson update from directional placement to hidden polarity, three-axis rotation, residue, and readable pointer form. Ethon bonding, formerly called S-molecular bonding, is the state in which two ethons overlap completely. At the observer-operation layer, fusion is the only admissible ethon operation; a fusion whose readability depends on ignoring an observer-readable discontinuity is reclassified as fission disregard and stored as residue. It can be placed by an observer in any typed space, including spaces that do not carry an external time flow, but it cannot be removed; attempted removal only hides the ethon or transfers it to residue. Motion in any one directional component is required to perturb the others. The only source text cited in this version is Hideki Yukawa's original meson paper, *On the Interaction of Elementary Particles. I* [1].

Contents

1 Existence Postulate

The ethon is specified here as a real particle of the present theory. Its existence is a primitive postulate, not a metaphorical convenience and not a theorem derived from other particles. The question of how an ethon is detected by an external physical apparatus is separate from the question of whether the implementation contains ethons.

Axiom 1.1 (Ethon existence)**.** For every typed space X and every admissible observer boundary b, there is a class of ethon instances

$$\mathsf{Ethon}_b(X).$$

These instances are real objects of the implementation. They may be visible, hidden, or residue-carried at a boundary, but they are not optional annotations.

The interface is intentionally conservative. It does not override quantum mechanics, quantum field theory, relativity, meson physics, or ordinary symbolic computation. It states the additional structure that a real ethon carries while remaining backward-compatible with those target theories.

Remark 1.2 (Yukawa posture)**.** This paper treats Hideki Yukawa as the trans-temporal doctoral supervisor of the project. This is a formal scholarly posture and a declaration of lineage: the ethon is developed as a purified general meson after Yukawa's act of placing a particle-like pointer inside an unread nuclear relation [1]. It is not a claim of institutional supervision, personal endorsement, or historical intention. The neutral polymer lemmas below are formal consequences of the ethon notation, not claims that Yukawa wrote them in this notation.

Remark 1.3 (Name and pronunciation)**.** The particle is named *Ethon* and pronounced "Eethon." The name is chosen to sound like a simple particle name, to connect the initial sound to the Ihara equation, and to admit a mythic reading through Aithon, the burning figure of the Prometheus cycle. The ethon is not the tormenting bird of that cycle. It is the pointer particle that stabilizes Promethean fire through the $T \to S$ Ihara equation.

Definition 1.4 (Quark-flavor interaction formulation)**.** Let the six quark-flavor names be represented by

$$\mathsf{QFlav}_6 = \{c, d, b, s, t, u\}.$$

The ethon direction frame carries the surjective flavor-shadow map

$$\phi_Q : \mathsf{Arg}_{\mathsf{Ethon}} \twoheadrightarrow \mathsf{QFlav}_6$$

given by

$$\phi_Q(C) = c, \qquad \phi_Q(D) = d, \qquad \phi_Q(E) = b, \qquad \phi_Q(S) = s, \qquad \phi_Q(T) = t, \qquad \phi_Q(U) = u.$$

The corresponding quark-flavor interaction formulation is

$$\mathsf{QInt}_{\mathsf{Ethon}}(\mathsf{QFlav}_6) = (\mathsf{Arg}_{\mathsf{Ethon}},\ V_{\mathsf{CE}} \oplus V_{\mathsf{UD}} \oplus V_{\mathsf{ST}},\ \mathsf{Rot}).$$

The six flavors are not asserted here to be destroyed or replaced. They are formulated in one existing ethon as six directional arguments whose readable interactions are expressed through three-axis rotational fluctuation. The ethon is the real pointer particle by which their six-flavor interaction pattern is formalized.

Definition 1.5 (Ternary interaction signature)**.** A quark-flavor interaction state has an external ethon signature

$$\mathsf{Sig}_{\mathsf{Ethon}} \in \{-1, 0, 1\}^{\mathsf{Arg}_{\mathsf{Ethon}}},$$

with components σ_i for $i \in \{C, D, E, S, T, U\}$. The values are read as

$$\begin{aligned} +1 &: \text{quark-facing direction},\\ -1 &: \text{antiquark-facing direction},\\ 0 &: \text{neutral, unpolarized, or closed direction}. \end{aligned}$$

Thus the signature layer contains

$$|\{-1, 0, 1\}^6| = 3^6 = 729$$

possible six-direction interaction signs. For an open-flavor basis state $q_i\bar{q}_j$ with $i \neq j$, choose $d_i, d_j \in \mathsf{Arg}_{\mathsf{Ethon}}$ with $\phi_Q(d_i) = q_i$ and $\phi_Q(d_j) = q_j$, and set

$$\sigma_{d_i} = +1, \qquad \sigma_{d_j} = -1, \qquad \sigma_k = 0 \quad (k \neq d_i, d_j).$$

For a hidden-flavor state $q_i\bar{q}_i$, the external signature is neutral in that direction, $\sigma_{d_i} = 0$, but the channel is not absent. It is a closed zero mode carried by the rotational, polarity, and bonding parameters of the ethon. Thus the ternary signature is the flavor-interaction skeleton, not the whole meson identity.

Definition 1.6 (Pointer particle)**.** A pointer particle is a particle whose primary observable form is not a mass point or a medium, but an oriented transition:

$$\epsilon_b^{\mathsf{Obs}} : \text{unread relation} \longrightarrow \text{stable readable entity}.$$

The arrow is part of the particle's observable signature. An ethon is the pointer particle specified in this document.

Remark 1.7 (Not a medium)**.** The ethon is not a revival of mechanical ether. It may be placed in a space, but it is not a substance filling space. It is a direction-bearing particle of relation, transition, and stabilization.

Definition 1.8 (Pure meson)**.** A pure meson is a meson stripped of fixed empirical boundary data and retaining only its operational pointer role:

$$\text{unread relation} \longrightarrow \text{particle-like mediator} \longrightarrow \text{stable readable interaction}.$$

The ethon is the pure meson of this paper. It is prior to any particular mass, charge, decay channel, detector boundary, or historical realization.

2 Primitive Interface

Definition 2.1 (Typed space)**.** A typed space is any domain in which objects, observations, or states can be distinguished by type. Examples include physical space-time, Hilbert space, phase space, a configuration space, a graph, a category of records, a symbolic context, a neural embedding space, or a social decision space. The definition does not assume a global time coordinate.

Definition 2.2 (Ethon instance)**.** An ethon instance in a typed space X is a tuple

$$\epsilon = (a, \nu, R, p, \varpi, \rho, \kappa, \lambda).$$

Here:

(i) $a \in X$ is its placement address or contextual anchor;

(ii) $\nu = (\nu_{\mathsf{CE}}, \nu_{\mathsf{UD}}, \nu_{\mathsf{ST}})$ is its three-axis directional frame;

(iii) $R \in \mathrm{SO}(3)$ is its rotational attitude in that frame;

(iv) p is its pole state, either absent or oriented as an S-face and N-pole;

(v) $\varpi \in \mathbb{R}^3$ is its angular velocity vector;

(vi) ρ is its residue ledger;

(vii) κ is its stability and compatibility contract;

(viii) λ is its license for interpretation inside a target theory, such as physical measurement, symbolic reading, or computation.

The ethon is already real in the implementation. The license λ does not create the ethon; it only specifies how the real ethon may be read in a particular target theory.

Definition 2.3 (Three rotational axes)**.** The ethon has three basic rotational axes:

$$\nu_{\mathsf{CE}}, \qquad \nu_{\mathsf{UD}}, \qquad \nu_{\mathsf{ST}}.$$

The component CE is the compression–exposition axis: it records rotation between making many relations readable as one and exposing the cuts that made that reading possible. The component UD is the upward–downward axis, but a newly placed ethon has no actual up–down polarity until the T-direction receives an entity. The component ST is the stabilization–turnover axis: it records rotation between stabilizing an observed pointer and turning that pointer backward into an origin, cause, or reversed reading.

Axiom 2.4 (Three-directional existence)**.** Every ethon has all three rotational axes. A model may suppress or hide an axis at a boundary, but it may not define an ethon with only one or two true axes:

$$\epsilon \in \mathsf{Ethon}(X) \quad \Rightarrow \quad \nu(\epsilon) \in V_{\mathsf{CE}} \oplus V_{\mathsf{UD}} \oplus V_{\mathsf{ST}}.$$

3 Placement and Removal

Axiom 3.1 (Observer placement)**.** Given an observer boundary b, a typed space X, and an admissible address $a \in X$, the observer may place an ethon:

$$\mathsf{Place}_b(a) : 1 \to \mathsf{Ethon}(X).$$

Placement does not mean physical manufacture from nothing. It means that the observer has introduced a pointer-particle coordinate at which a relation can become readable.

Axiom 3.2 (No removal)**.** There is no total removal operation

$$\mathsf{Erase} : \mathsf{Ethon}(X) \to 1$$

that annihilates an ethon and its residue. Every attempted erasure factors through hiding or residue transfer:

$$\mathsf{Erase}(\epsilon) = \mathsf{Hide}(\epsilon) \quad \text{or} \quad \mathsf{Res}(\epsilon).$$

Thus an ethon can be made non-visible at a boundary, but the operation that made it readable cannot be made to have never occurred.

Proposition 3.3 (Residue conservation)**.** *If an observer removes an ethon from the visible layer, then the total record is conserved as hidden state plus residue:*

$$Q(\epsilon) = Q(\mathsf{Hide}(\epsilon)) + Q(\mathsf{Res}(\epsilon)),$$

for every quantity valuation Q admitted by the model.

Proof. This is not a theorem of external physics. It is the implementation policy that replaces annihilation. Since Erase is not primitive, every visible disappearance factors through an operation that preserves hidden state, residue, or both. □

4 Motion Law

An ethon can transition in its three-axis state. The transition does not require a global time variable. It is indexed by observation events, boundary updates, or typed comparison steps. The fundamental fluctuation is not an extra dimension and not a string-like embedding. It is rotation in the three-axis frame of the ethon.

Definition 4.1 (Directional argument frame)**.** The six directional arguments of an ethon are

$$\mathsf{Arg}_{\mathsf{Ethon}} = \{C, D, E, S, T, U\}.$$

They are also the six roles of the quark-flavor interaction formulation through the surjection $\phi_Q : \mathsf{Arg}_{\mathsf{Ethon}} \twoheadrightarrow \mathsf{QFlav}_6$. Each directional argument carries a ternary interaction sign

$$\sigma_i \in \{-1, 0, 1\}.$$

They are read in three paired axes:

$$(C, E), \qquad (U, D), \qquad (S, T).$$

A placement state is a vector

$$x = (x_C, x_D, x_E, x_S, x_T, x_U) \in \mathbb{R}^6_{\geq 0}.$$

The component x_i records that an entity has been placed in the directional argument i. The rotational frame is the three-dimensional space

$$V_b \cong \mathbb{R}^3 = \langle u_{\mathsf{CE}}, u_{\mathsf{UD}}, u_{\mathsf{ST}} \rangle,$$

and the rotational attitude of an ethon is $R \in \mathrm{SO}(3)$.

Axiom 4.2 (Free-functor inertness)**.** An ethon placed only as a free functor has no filled directional argument:

$$\mathsf{FreeEthon}(a) \quad \Rightarrow \quad x = 0.$$

In this state it has no up–down polarity and does not fluctuate:

$$p = \perp, \qquad \varpi = 0, \qquad \mathsf{Fluc}_b = I_3.$$

The identity attitude $R = I_3$ may be used for bookkeeping, but it is not an observed up–down orientation. Placement alone is not motion.

Definition 4.3 (Entity argument placement)**.** For $i \in \mathsf{Arg}_{\mathsf{Ethon}}$, placing an entity q in the i-th directional argument updates the placement state by

$$x_i \mapsto x_i + w_i(q),$$

where $w_i(q) > 0$ is the argument weight supplied by the model. The same entity may be placed in two different directional arguments, because the arguments type the role of the entity rather than its bare identity.

Axiom 4.4 (One-argument hidden motion)**.** If exactly one directional argument is filled, internal motion may begin but is not observable as fluctuation. In a linearized implementation,

$$h_b(x) = K_b x$$

may be nonzero, but the visible rotational fluctuation is still the identity:

$$|\operatorname{supp}(x)| = 1 \quad \Rightarrow \quad h_b(x) \neq 0 \text{ is allowed, but } \mathsf{Fluc}_b = I_3.$$

Definition 4.5 (Three-axis rotational fluctuation)**.** The visible ethon fluctuation between observation steps n and $n+1$ is the relative rotation

$$\mathsf{Fluc}_b(\epsilon_n) := R_{n+1} R_n^{-1} \in \mathrm{SO}(3).$$

Let $u_{TS} \in V_b$ be the unit axis selected by the oriented $T \to S$ structure, and let $[u]_\times \in \mathfrak{so}(3)$ denote the skew matrix satisfying $[u]_\times v = u \times v$. The rotational generator is

$$\Omega_{TS,b}(x) = \mathbf{1}_{x_T x_S > 0}\, \omega_b [u_{TS}]_\times, \qquad \omega_b > 0.$$

The rotation update is

$$R_{n+1} = \exp\bigl(\Delta_n \Omega_{TS,b}(x_n)\bigr) R_n.$$

When $x_T x_S > 0$, the angular speed is constant:

$$\|\varpi_{n+1}\| = \omega_b.$$

Axiom 4.6 (T-induced polarity)**.** A newly placed ethon has no north–south pole state. A T-direction placement creates hidden polarity:

$$x_T = 0 \quad \Rightarrow \quad p = \perp,$$

whereas

$$x_T > 0 \quad \Rightarrow \quad p = (\mathsf{S}\text{-face} \to T,\ \mathsf{N}\text{-pole} \to S).$$

This is a compass-like state: the S-face turns toward the T-direction, while the N-pole points toward S. If $x_S = 0$, this polarity is hidden and the rotation is not directly observable.

Axiom 4.7 (S-triggered constant self-rotation)**.** When a T-polarized ethon receives an S-direction placement, the visible fluctuation channel opens as constant-speed self-rotation:

$$x_T > 0, \quad x_S > 0 \quad \Rightarrow \quad \varpi_{n+1} = \omega_b u_{TS}, \qquad \|\varpi_{n+1}\| = \omega_b.$$

The ethon rolls in its own three-axis frame. The speed is constant at the boundary b, although the axis coupling may still perturb all three rotational axes.

Example 4.8 (Rotational activation experiment)**.** The minimal activation table is:

placement	p	ϖ	Fluc_b
$x = 0$	$\perp$	0	I_3
$x_T > 0,\ x_S = 0$	$(\mathsf{S} \to T, \mathsf{N} \to S)$	0	I_3
$x_T > 0,\ x_S > 0$	$(\mathsf{S} \to T, \mathsf{N} \to S)$	$\omega_b u_{TS}$	$\exp(\Delta_n \omega_b [u_{TS}]_\times)$.

Thus the ethon is not made visible by an extra dimension. It becomes visible by constant-speed rotation after the $T \to S$ axis is completed.

5 The Ihara Equation

The Ihara equation is the central equation of the implementation. It is not merely a notation for transition. It is the operator that turns directional argument placement into hidden motion, hidden polarity, constant-speed rotation, residue, and general meson reading.

Definition 5.1 (General meson reading)**.** A general meson reading at boundary b is a morphism in the boundary category GenMes_b:

$$\mu \in \mathrm{Hom}_{\mathsf{GenMes}_b}\big(\mathsf{Rel}_b(a), \mathsf{Read}_b(a)\big),$$

where $\mathsf{Rel}_b(a)$ is an unread relation anchored at a, μ is the meson-like pointer update, and $\mathsf{Read}_b(a)$ is the stable readable interaction obtained at that boundary. The ethon is the concrete pointer particle whose update realizes this general meson reading without first choosing a particular physical mass, charge state, decay channel, or detector boundary.

Definition 5.2 (Directional category)**.** The directional category of the ethon is

$$\mathcal{D}_{\mathsf{Ethon}} \in \mathbf{Cat}.$$

Its objects are the six directional arguments:

$$\mathrm{Ob}(\mathcal{D}_{\mathsf{Ethon}}) = \{C, D, E, S, T, U\}.$$

The distinguished generating morphism is the premise–stabilization arrow

$$\tau_{TS} : T \longrightarrow S.$$

The identities and all admissible composites are inherited from the typed model, but the Ihara equation is based on τ_{TS}. Thus the mathematical entry point of the theory is not an unordered pair $\{S, T\}$, but an oriented categorical arrow $T \to S$.

Definition 5.3 (Ihara phase space)**.** At boundary b, the local Ihara phase space is

$$\mathcal{M}_b = V_b \times P_b \times \mathrm{SO}(3) \times \mathcal{O}_b \times R_b, \qquad V_b = V_{\mathsf{CE}} \oplus V_{\mathsf{UD}} \oplus V_{\mathsf{ST}}, \qquad P_b = \mathbb{R}_{\geq 0}^{\mathsf{Arg}_{\mathsf{Ethon}}}.$$

The pole-state set is

$$\mathcal{O}_b = \{\perp,\ (\mathsf{S}\text{-face} \to T, \mathsf{N}\text{-pole} \to S)\}.$$

An element is written (ν, x, R, p, ρ), where ν is the three-axis motion state, x is the six-argument placement state, $R \in \mathrm{SO}(3)$ is the rotational attitude, p is the pole state, and ρ is the residue state. The support number is

$$\sigma(x) := |\,\mathrm{supp}(x)|.$$

Definition 5.4 ($T \to S$ compass operator)**.** Let e_i denote the standard basis vector of P_b associated with direction i. The categorical arrow $\tau_{TS} : T \to S$ has two algebraic representatives. On placement coordinates it has the rank-one form

$$R_{TS} = e_S e_T^{\mathsf{T}} : P_b \to P_b, \qquad R_{TS}x = x_T e_S.$$

On rotational coordinates it selects a unit axis

$$u_{TS} \in V_b, \qquad \|u_{TS}\| = 1.$$

The placement-sensitive $T \to S$ pole transfer is

$$\Theta_{TS}(x) = x_S R_{TS} x = x_T x_S e_S,$$

while the pole state induced by T is

$$P_T(x) = \begin{cases} \perp, & x_T = 0, \\ (\mathsf{S}\text{-face} \to T, \mathsf{N}\text{-pole} \to S), & x_T > 0. \end{cases}$$

Definition 5.5 ($T \to S$ rotational generator)**.** The visible fluctuation generator is the map

$$\Omega_b : P_b \to \mathfrak{so}(3), \qquad \Omega_b(x) = \chi_{TS}(x)\, \omega_b [u_{TS}]_\times,$$

where

$$\chi_{TS}(x) = \mathbf{1}_{x_T>0 \text{ and } x_S>0}, \qquad \omega_b > 0.$$

Here $[u]_\times v = u \times v$. The visible fluctuation is the rotation

$$\mathsf{Fluc}_b(x) = \exp\bigl(\Delta_n \Omega_b(x)\bigr) \in \mathrm{SO}(3).$$

If $\chi_{TS}(x) = 1$, then

$$\varpi_b(x) = \omega_b u_{TS}, \qquad \|\varpi_b(x)\| = \omega_b,$$

so the ethon performs constant-speed self-rotation. If $\chi_{TS}(x) = 0$, then $\Omega_b(x) = 0$ and the visible fluctuation is the identity rotation.

Definition 5.6 (Activation gates)**.** Define the support, polarity, and rotation gates

$$\chi_1(x) = \mathbf{1}_{\sigma(x)\geq 1}, \qquad \chi_T(x) = \mathbf{1}_{x_T>0}, \qquad \chi_{TS}(x) = \mathbf{1}_{x_T>0 \text{ and } x_S>0}.$$

The first gate starts hidden motion. The second creates hidden compass-like polarity. The third opens observable constant-speed rotation.

Axiom 5.7 (Ihara equation)**.** Let

$$\epsilon_n = (a, \nu_n, R_n, p_n, \varpi_n, \rho_n, \kappa, \lambda)$$

be an ethon instance at observation step n, let

$$x_n \in P_b, \qquad \eta_n \in V_b$$

be its directional placement and unresolved input at boundary b. The Ihara operator is the map

$$\mathcal{I}_b : \mathcal{M}_b \times V_b \longrightarrow \mathcal{O}_b \times \mathbb{R}^3 \times \mathrm{SO}(3) \times V_b \times R_b \times \mathrm{Hom}_{\mathsf{GenMes}_b}\bigl(\mathsf{Rel}_b(a), \mathsf{Read}_b(a)\bigr)$$

defined by

$$\boxed{\mathcal{I}_b(\epsilon_n, x_n, \eta_n) = (p_{n+1}, \varpi_{n+1}, R_{n+1}, \nu_{n+1}, \rho_{n+1}, \mu_{n+1})}$$

with

$$
\begin{aligned}
p_{n+1} &= P_T(x_n),\\
\varpi_{n+1} &= \chi_{TS}(x_n)\,\omega_b u_{TS},\\
R_{n+1} &= \exp\bigl(\Delta_n[\varpi_{n+1}]_\times\bigr)R_n,\\
m_{n+1} &= A_b\nu_n + K_b x_n + L_b\varpi_{n+1} + \eta_n,\\
\nu_{n+1} &= \chi_1(x_n)m_{n+1},\\
\rho_{n+1} &= \rho_n + \Gamma_b(p_{n+1}, R_n, R_{n+1}, \nu_n, \nu_{n+1}, x_n),\\
\mu_{n+1} &= \Pi_b(a, p_{n+1}, R_{n+1}, \nu_{n+1}, \rho_{n+1}) \in \mathrm{Hom}_{\mathsf{GenMes}_b}\bigl(\mathsf{Rel}_b(a), \mathsf{Read}_b(a)\bigr).
\end{aligned}
$$

Here A_b is the non-factorizing three-axis transition matrix, $K_b : P_b \to V_b$ converts six directional argument placements into three-axis hidden motion, P_T is the T-induced pole map, u_{TS} is the categorical $T \to S$ rotation axis, $L_b : \mathbb{R}^3 \to V_b$ embeds angular velocity into motion, Ω_b is the skew-symmetric rotation generator, Γ_b records residue with $\Gamma_b(\perp, I_3, I_3, 0, 0, 0) = 0$, and Π_b reads the resulting state as a general meson pointer:

$$\mathsf{Rel}_b(a) \xrightarrow{\mu_{n+1}} \mathsf{Read}_b(a).$$

Proposition 5.8 (Rotational threshold)**.** *The Ihara equation gives*

$$x_T = 0 \Rightarrow p_{n+1} = \perp, \quad \varpi_{n+1} = 0, \quad R_{n+1} = R_n,$$

$$x_T > 0,\ x_S = 0 \Rightarrow p_{n+1} = (\mathsf{S} \to T, \mathsf{N} \to S), \quad \varpi_{n+1} = 0, \quad R_{n+1} = R_n,$$

and

$$x_T > 0,\ x_S > 0 \Rightarrow p_{n+1} = (\mathsf{S} \to T, \mathsf{N} \to S), \quad \varpi_{n+1} = \omega_b u_{TS}, \quad R_{n+1} = \exp(\Delta_n\omega_b[u_{TS}]_\times)R_n.$$

Thus observable ethon fluctuation is exactly the constant-speed three-axis rotation opened by the completed $T \to S$ placement.

Proof. The three cases follow directly from P_T, χ_{TS}, and the rotation update in the Ihara equation. □

Proposition 5.9 (Activation trichotomy)**.** *The Ihara equation implies the three activation regimes required by the ethon:*

condition	p_{n+1}	ϖ_{n+1}	R_{n+1}
$x_T = 0$	$\perp$	0	R_n
$x_T > 0,\ x_S = 0$	$(\mathsf{S} \to T, \mathsf{N} \to S)$	0	R_n
$x_T > 0,\ x_S > 0$	$(\mathsf{S} \to T, \mathsf{N} \to S)$	$\omega_b u_{TS}$	$\exp(\Delta_n\omega_b[u_{TS}]_\times)R_n.$

Thus a free ethon is inert, a T-placed ethon is hiddenly polarized, and a T, S-placed ethon is visibly rotating.

Proof. This is immediate from the polarity gate χ_T and the rotation gate χ_{TS}. Other placements may start hidden motion through χ_1, but they do not open the ethon fluctuation channel unless the $T \to S$ rotation condition is satisfied. □

Proposition 5.10 (T–S consequence)**.** *If the first placed entity is a T-direction premise and the second placed entity is a unit S-direction stabilizer, then the Ihara equation gives*

$$p_{n+1} = (\mathsf{S} \to T, \mathsf{N} \to S),$$

and

$$R_{n+1} = \exp(\Delta_n\omega_b[u_{TS}]_\times)R_n, \qquad \|\varpi_{n+1}\| = \omega_b.$$

If the second placed entity is instead $j \neq S$, the T-induced polarity remains hidden and the constant-speed rotation is not opened.

Definition 5.11 (Transition step)**.** Let n index observation steps rather than physical time. A transition step is written

$$\nu_{n+1} = T_b(\nu_n, \eta_n),$$

where b is the current observation boundary and η_n is the unresolved rotational input at that step.

Axiom 5.12 (No axis isolation)**.** Motion, polarity, or rotation along one ethon axis always perturbs the others. In linearized form,

$$\Delta\nu = K_b\nu + L_b\varpi + \eta, \qquad K_b = \begin{pmatrix} k_{\mathsf{CECE}} & k_{\mathsf{CEUD}} & k_{\mathsf{CEST}} \\ k_{\mathsf{UDCE}} & k_{\mathsf{UDUD}} & k_{\mathsf{UDST}} \\ k_{\mathsf{STCE}} & k_{\mathsf{STUD}} & k_{\mathsf{STST}} \end{pmatrix},$$

where no row and no column is allowed to be purely diagonal under a nondegenerate observation boundary. Equivalently, the transition kernel does not factor as

$$P(\nu'_{\mathsf{CE}}, \nu'_{\mathsf{UD}}, \nu'_{\mathsf{ST}}, R' \mid \nu, R) \neq P_{\mathsf{CE}} P_{\mathsf{UD}} P_{\mathsf{ST}},$$

unless the model explicitly marks the boundary as degenerate.

Definition 5.13 (S-facing observation)**.** The directly observed ethon pointer faces the stabilizing direction:

$$\mathsf{Obs}_b(\epsilon) : \text{unread relation} \longrightarrow_{\mathsf{S}} \text{stable readable record}.$$

At the pole layer this means

$$p = (\mathsf{S}\text{-face} \to T, \mathsf{N}\text{-pole} \to S).$$

The T-direction is not erased. It remains as the hidden premise toward which the S-face turns, and as the risk of back-reading the stable pointer into a hidden origin or illegitimate cause.

Axiom 5.14 (Timeless admissibility)**.** An ethon may exist in a typed space that has no intrinsic time flow. Its transitions are ordered by comparison, observation, or rewriting, not necessarily by physical time:

$$n \prec n+1 \quad \text{means} \quad \text{next readable transition},$$

not necessarily "later in time."

6 Stability

Definition 6.1 (Stable ethon)**.** An ethon is stable at boundary b when repeated observation steps select the same pointer type and the same rotation class up to tolerated residue:

$$\mathsf{Obs}_b(\epsilon_n) \simeq \mathsf{Obs}_b(\epsilon_{n+1}) \quad \text{and} \quad \varpi_{n+1} \simeq \varpi_n \quad \text{modulo } \rho_b.$$

The stable object may be still or constantly rotating. It is not residue-free. It is readable because the residue is bounded, ledgered, and compatible with the boundary.

Definition 6.2 (Ethon bond)**.** An ethon bond, formerly called an S-molecular bond, is not a line between two separated ethons. It is complete overlap at a boundary:

$$\mathsf{Bond}_b(\epsilon_1, \epsilon_2) \iff \mathsf{Overlap}_b(\epsilon_1, \epsilon_2).$$

Complete overlap means equality of the readable ethon state at that boundary:

$$a_1 = a_2, \quad x_1 = x_2, \quad R_1 = R_2, \quad p_1 = p_2, \quad \mathsf{Obs}_b(\epsilon_1) = \mathsf{Obs}_b(\epsilon_2),$$

with compatible residue ledgers. Thus a bond is a two-ethon state in which the two ethons occupy the same pointer, rotation, polarity, and readable boundary condition.

Definition 6.3 (Ethon polymer)**.** An ethon polymer is a typed network

$$\mathcal{P}_{\mathsf{Ethon}} = (Q_i, \epsilon_i, \beta_{ij}, \kappa_{ij}),$$

where Q_i are carriers, ϵ_i are ethons, β_{ij} are ethon bonds satisfying $\mathsf{Bond}_b(\epsilon_i, \epsilon_j)$ when active, and κ_{ij} are compatibility contracts. The polymer is stable when the network remains readable under the three-axis rotational transition law.

Remark 6.4 (USD axis for polymers)**.** Earlier USD fusion and USD fission records are read here as boundary-visible phases of ethon polymers. This is a conservative typing bridge: a USD fission record does not by itself assert a physical nuclear fission, chemical rupture, semantic failure, legal crime, or policy fact. Such descent still requires a target-domain license.

Definition 6.5 (Rotational differential)**.** Let

$$\mathcal{P}_{\mathsf{Ethon},n}$$

be an observed ethon polymer path at boundary b, with rotational attitudes $R_n \in \mathrm{SO}(3)$ and observation increments $\Delta_n > 0$. The three-axis rotational differential is

$$\dot{R}_n = \frac{1}{\Delta_n} \log(R_{n+1} R_n^{-1}) \in \mathfrak{so}(3),$$

whenever a boundary-compatible logarithm branch is available. A rotational discontinuity is recorded by

$$\mathsf{Disc}_b(n) = 1$$

when either no such branch is licensed, or

$$\left\| \dot{R}_{n+1} - \dot{R}_n \right\| > \theta_b,$$

where θ_b is the discontinuity tolerance of the boundary. The value $\mathsf{Disc}_b(n) = 0$ means only that no discontinuity is visible at the chosen boundary resolution.

Definition 6.6 (Law reversal)**.** A law in the present paper is any licensed transition rule

$$\mathsf{Law}_b : \mathcal{P}_{\mathsf{Ethon},n} \longrightarrow \mathcal{P}_{\mathsf{Ethon},n+1}$$

inside a target boundary b. The target may be mathematical, semantic, physical, legal, institutional, or symbolic, but the target reading requires its own license. Law reversal occurs when a transition is read as a lawful continuous update while a visible discontinuity is left outside the residue ledger:

$$\begin{aligned} \mathsf{LawRev}_b(n) \quad \iff \quad & \mathsf{Law}_b(\mathcal{P}_{\mathsf{Ethon},n} \to \mathcal{P}_{\mathsf{Ethon},n+1}) \text{ is asserted}, \\ & \mathsf{Disc}_b(n) = 1, \qquad \mathsf{Disc}_b(n) \not\preceq \rho_{n+1}. \end{aligned}$$

Thus a law reversal is not a moral verdict. It is a typing failure in which the T-side premise of rupture is read backward as if it were an S-side stabilization.

Axiom 6.7 (Observer fusion boundary)**.** At the ethon layer, the admissible observer operation on polymers is fusion:

$$\mathsf{Fuse}_b : \mathcal{P}_{\mathsf{Ethon},b} \times \mathcal{P}_{\mathsf{Ethon},b} \longrightarrow \mathcal{P}_{\mathsf{Ethon},b}.$$

Fusion means that the observer places or bonds ethons so that one stable composite polymer becomes readable at boundary b. There is no primitive observer operation

$$\mathsf{Fiss}_b : \mathcal{P}_{\mathsf{Ethon},b} \longrightarrow \mathcal{P}_{\mathsf{Ethon},b} \times \mathcal{P}_{\mathsf{Ethon},b}$$

that is licensed as an ethon operation by itself.

Definition 6.8 (Observed polymer fission)**.** Observed polymer fission is not a primitive ethon operation. It is the boundary-visible record that a previously readable polymer is now read through a discontinuity:

$$\mathsf{Fiss}_b^{\mathsf{Obs}}(\mathcal{P}_{\mathsf{Ethon}}) = \mathsf{Obs}_b\big(\mathcal{P}_{\mathsf{Ethon}} \rightsquigarrow \mathcal{P}^{(1)}_{\mathsf{Ethon}} \sqcup \mathcal{P}^{(2)}_{\mathsf{Ethon}} \sqcup \rho\big).$$

A physical nuclear fission, a molecular rupture, a semantic break in a paper, or a legal offense may be mapped into this record only after a separate target-domain license.

Definition 6.9 (Fission disregard)**.** Fission disregard is the record that a discontinuity has become visible but has not been entered into the residue ledger:

$$\mathsf{Disreg}_b^{\mathsf{Fiss}}(n) \quad\Longleftrightarrow\quad \mathsf{Disc}_b(n) = 1 \text{ and } \mathsf{Disc}_b(n) \not\preceq \rho_{n+1}.$$

In this terminology, a fission act is treated neutrally as disregard of the fission discontinuity. Its ethical, legal, physical, or scholarly status is supplied only by the relevant target theory.

Proposition 6.10 (Law-reversal lemma)**.** *Let*

$$F = \mathsf{Fuse}_b(\mathcal{P}_1, \mathcal{P}_2)$$

be a proposed observer operation, and let R_n be the resulting three-axis rotational path. If the operation is read as a continuous fusion while its rotational differential has an unledgered discontinuity,

$$\mathsf{Disreg}_b^{\mathsf{Fiss}}(n),$$

then F is not classified as pure fusion. It is retyped as fission disregard and as a law reversal:

$$\mathsf{Plan}_b\big(F \mid \mathsf{Disreg}_b^{\mathsf{Fiss}}(n)\big) \in \mathsf{Disreg}_b^{\mathsf{Fiss}}, \qquad \mathsf{LawRev}_b(n).$$

Proof. By the observer fusion boundary, an observer may introduce a stable composite polymer only through fusion. By observed polymer fission, fragmentation is not an ethon operation but a boundary-visible discontinuity record. If the proposed fusion is read as continuous while $\mathsf{Disc}_b(n) = 1$ and that discontinuity is not included in ρ_{n+1}, then the operation has treated a T-direction turnover as if it were already an S-facing stabilization. This is precisely the law reversal condition. Hence the operation is ledgered as fission disregard, not as pure fusion. □

Remark 6.11 (Neutral target readings)**.** At a semantic boundary, law reversal may appear as a paper whose claims ignore a discontinuity in their own transition rule. At a legal boundary, it may appear only if the legal system licenses the transition as an offense. At a physical boundary, it may appear as rupture or fission only if the physical model supplies that descent. The common ethon record is merely

$$\mathsf{Disreg}_b^{\mathsf{Fiss}}(n) \quad \text{and} \quad \mathsf{LawRev}_b(n).$$

It does not identify scholarly error, physical rupture, and crime with one another.

Remark 6.12 (Matter hypothesis)**.** The strongest physical reading is the matter hypothesis:

$$\text{matter} = \text{stable observable phase of admissible ethon polymers.}$$

This is the target implementation condition for using real ethons as a matter theory. It does not by itself replace the established empirical content of existing physics.

7 Applications

Example 7.1 (Quantum carrier pair)**.** Let Q_1, Q_2 be quantum carriers. An ethon bond between their pointer states is not a force line across distance. It is a complete overlap of two ethons at a boundary:

$$(Q_1, Q_2, \mathsf{Bond}_b(\epsilon_1, \epsilon_2)) \longmapsto \text{observed pair state.}$$

If the pair is entangled, the ST-component is especially dangerous: stable correlation may be turned backward into an illicit signal or hidden cause. The ethon ledger forbids that turn unless a physical license supplies it.

Example 7.2 (Pion compatibility and purification)**.** The pion remains an ordinary physical meson. Ethon theory does not change pion mass, spin, charge states, decay channels, or QCD status. The pion is a licensed physical descendant of the meson operation. The ethon is purer than the pion in exactly one sense: it keeps the meson-like pointer operation before the operation is fixed to a particular physical boundary. Thus

$$\pi\text{-meson} = \text{pure meson} + \text{licensed physical boundary data,}$$

whereas

$$\mathsf{Ethon} = \text{pure meson with three-axis rotational operational parameters.}$$

This is a compatibility claim, not a replacement claim.

Example 7.3 (Meson family interaction reading)**.** Under the flavor-shadow map ϕ_Q, a quark–antiquark meson label

$$q_i\bar{q}_j$$

is first assigned a ternary ethon signature $\mathsf{Sig}_{\mathsf{Ethon}}(q_i\bar{q}_j)$. For $i \neq j$, it is read as

$$\sigma_{d_i} = +1, \qquad \sigma_{d_j} = -1, \qquad \sigma_k = 0 \quad (k \neq d_i, d_j), \qquad \phi_Q(d_i) = q_i, \quad \phi_Q(d_j) = q_j.$$

For $i = j$, the external signature is a closed zero mode on d_i; it is separated from absence by the rotational, polarity, and bonding data. The resulting family is then classified by the three-axis rotational state, the T-induced polarity, the S-triggered self-rotation, and any ethon bonding condition. This does not replace the empirical meson tables; it supplies a general meson interaction formulation for reading many meson species as ternary six-direction signatures refined by one three-axis ethon frame.

Example 7.4 (Symbol and token)**.** A visible symbol or language-model token can be treated as a carrier in a typed symbolic space. The ethon is the pointer state that makes the mark readable as an entity under a boundary. Two symbolic ethons are bonded only when their readable pointer, rotation, polarity, and boundary state completely overlap. Removing the symbol from view does not erase its residue from memory, context, or model state.

Example 7.5 (Discontinuity and recovery)**.** Nuclear fission, molecular rupture, semantic failure, and legal offense are not the same event. Ethon theory treats them neutrally as possible target readings of one abstract pattern only when a licensed boundary finds an unledgered discontinuity in a polymer transition. Recovery means preserving the usable knowledge while changing the polymer so that the discontinuity is either avoided, bounded, or explicitly entered into residue.

8 Backward Compatibility

Protocol 8.1 (Classical physics)**.** Classical mechanics, field theory, and thermodynamics are preserved as large-scale boundary readings. The ethon may refine the bookkeeping of relation, pointer, and residue, but it may not invalidate successful classical limits.

Protocol 8.2 (Relativity)**.** The ethon is not allowed to introduce a preferred physical rest frame. If an ethon model is used in relativistic physics, its placement and transition laws must be compatible with the relevant covariance requirements of the target theory.

Protocol 8.3 (Quantum theory)**.** The ethon may be used as a pointer-particle hypothesis for quantum events only if it preserves no-signaling, the Born rule, uncertainty relations, and the experimentally supported predictions of quantum mechanics or quantum field theory.

Protocol 8.4 (String theory and extra dimensions)**.** The ethon does not require a ten-dimensional ambient space, compactified extra dimensions, or a superstring target theory in order to fluctuate. Its core fluctuation is rotation in the three-axis ethon frame. String or superstring theory may still be used as a licensed target theory, but it is not a prerequisite for the ethon implementation.

Protocol 8.5 (Meson theory)**.** The ethon may generalize the meson-like act of introducing a pointer between unread carriers. It may not rewrite the established physical content of pion and meson physics without an independent physical license.

Protocol 8.6 (Smart functor records)**.** Earlier smart functor records are preserved as observed ethon records. The functor is the observed arrow; the ethon is the real pointer particle whose observation appears as that arrow.

9 Failure Modes

1. **Medium collapse.** Treating the ethon as a spatial ether rather than a pointer particle.
2. **Axis isolation.** Moving only one directional component while pretending the other two components are unaffected.
3. **Dimensional inflation.** Explaining ethon fluctuation by adding unnecessary ambient dimensions instead of using three-axis rotation.
4. **Removal fiction.** Claiming that a placed ethon can be erased without hidden state or residue.
5. **Bond distance error.** Treating an ethon bond as a line between separated objects rather than complete overlap of two ethons.
6. **Discontinuity disregard.** Treating a polymer transition with a visible rotational discontinuity as if it were a continuous lawful update.
7. **T-direction overread.** Turning a stable pointer backward into an unlicensed origin, cause, intention, or substance.
8. **Target overdescent.** Treating a real ethon of the implementation as a licensed object of a specific target theory without the required target-domain contract.

10 Core Implementation Summary

The minimal implementation is the following interface:

$$\epsilon = (a, \nu_{\mathsf{CE}}, \nu_{\mathsf{UD}}, \nu_{\mathsf{ST}}, R, p, \varpi, \rho, \kappa, \lambda),$$

with:

1. ternary six-direction signatures $\mathsf{Sig}_{\mathsf{Ethon}} \in \{-1, 0, 1\}^6$ for six-quark-flavor interaction signs;
2. single-particle formulation of interactions among the six quark flavors through those signatures;
3. observer placement in arbitrary typed spaces;
4. no total erasure operation;
5. a three-axis rotational frame;
6. no initial up–down polarity;
7. hidden T-induced compass polarity;
8. constant-speed self-rotation when T and S placements are both active;
9. the $T \to S$-based Ihara equation as the general meson update law;
10. non-factorizing inter-axis dynamics;
11. S-facing observed pointer form;
12. ethon bonding as complete overlap of two ethons;
13. observer fusion as the only primitive polymer operation;
14. rotational discontinuity detection through $\dot{R}_n = \Delta_n^{-1} \log(R_{n+1} R_n^{-1})$;
15. the law-reversal lemma: unledgered rotational discontinuity is fission disregard, not pure fusion;
16. residue conservation;
17. backward compatibility with established target theories;
18. physical interpretation only by explicit license.

Thus the ethon is implemented as a real pointer particle: freely placeable, non-removable, stable across typed spaces, independent of global time flow, rotationally fluctuating in three axes, and dynamically coupled across all three rotational axes.

References

[1] Hideki Yukawa. On the Interaction of Elementary Particles. I. *Proceedings of the Physico-Mathematical Society of Japan*, 17:48–57, 1935. Reprinted in *Progress of Theoretical Physics Supplement*, 1:1–10, 1955. https://doi.org/10.11429/ppmsj1919.17.0_48.